INEQUALITIES

- **Elementary Results**:

(1) If $a > b$ and $b > c$, then $a > c$.
(2) In general if $a_1 > a_2, a_2 > a_3, a_3 > a_4, ..., a_{n-1} > a_n$, then $a_1 > a_n$.
(3) If $a > b$, then $a \pm c > b \pm c$.
(4) If $a > b$ and $c > 0$, then $ac > bc$.
(5) If $a > b$ and $c < 0$, then $ac < bc$.
(6) If $a > b$ and $c > 0$, then $\frac{a}{c} > \frac{b}{c}$.
(7) If $a > b$ and $c < 0$, then $\frac{a}{c} < \frac{b}{c}$.
(8) If $a > b > 0$, then $\frac{1}{a} < \frac{1}{b}$.
(9) If $a_i > b_i > 0,\ i = 1,2,...,n$, then $a_1 a_2 a_3 a_4 ... a_n > b_1 b_2 b_3 b_4 ... b_n$.
(10) If $0 < a < 1$ and r is positive real number, then $0 < a^r < 1 < \frac{1}{a^r}$.
(11) If $a > 1$ and r is positive real number, then $a^r > 1 > \frac{1}{a^r}$.
(12) If $0 < a < b$ and r is positive real number, then $a^r < b^r$.
(13) If $0 < a < b$ and r is positive real number, then $\frac{1}{a^r} > \frac{1}{b^r}$.
(14) If $0 < \frac{a}{b} < 1$ and r is positive real number, then $\left(\frac{a}{b}\right)^r < 1 < \left(\frac{b}{a}\right)^r$.
(15) Triangle Inequality: $|a + b| \le |a| + |b|$.
(16) $|a_1 + a_2 + a_3 + ... + a_n| \le |a_1| + |a_2| + |a_3| + ... + |a_n|$.
(17) $|a| = \max\{a, -a\}$.
(18) $-|a| \le a \le |a|, \qquad \forall a \in R$.
(19) If $b \ge 0$, then $|x - a| \le b \Rightarrow a - b \le x \le a + b$.
(20) If $a > 1$ and $x > y > 0$, then $\log_a x > \log_a y$.
(21) If $0 < a < 1$ and $x > y > 0$, then $\log_a x < \log_a y$.

- **Relation between AM (A), GM (G) and HM (H):** Let $a_1, a_2, a_3, a_4, ..., a_n$ are n number of positive real numbers then,

(1) $A = \frac{a_1 + a_2 + a_3 + ... + a_n}{n}, \quad G = (a_1 a_2 a_3 ... a_n)^{1/n}, \quad H = \frac{n}{\frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} + ... + \frac{1}{a_n}}$.
(2) $A \ge G \ge H$.
(3) A, G and H lie between the least and the greatest between $a_1, a_2, a_3, a_4, ..., a_n$.

- For n positive real numbers $a_1, a_2, a_3, ..., a_n$ and n positive real numbers $w_1, w_2, w_3, ..., w_n$, then

  (1) Arithmetic Mean = $A_w = \dfrac{w_1 a_1 + w_2 a_2 + w_3 a_3 + ... + w_n a_n}{w_1 + w_2 + w_3 + ... + w_n}$.

  (2) Geometric Mean = $G_w = \left(a_1^{w_1} a_2^{w_2} a_3^{w_3} ... a_n^{w_n}\right)^{\frac{1}{w_1 + w_2 + w_3 + ... + w_n}}$.

  (3) $A_w \geq G_w$.

  (4) $A_w = G_w$ iff $a_1 = a_2 = a_3 = ... = a_n$.

- **<u>Cauchy-Schwarz Inequality</u>**: For real numbers $a_1, a_2, a_3, ..., a_n$ and $b_1, b_2, b_3, ..., b_n$,
  $(a_1 b_1 + a_2 b_2 + a_3 b_3 ... + a_n b_n)^2 \leq \left(a_1^2 + a_2^2 + a_3^2 + ... + a_n^2\right)\left(b_1^2 + b_2^2 + b_3^2 + ... + b_n^2\right)$.
  The inequality hold iff $\dfrac{a_1}{b_1} = \dfrac{a_2}{b_2} = \dfrac{a_3}{b_3} = ... = \dfrac{a_n}{b_n}$.

- **<u>Tchebychef Inequality</u>**: For real numbers $a_1, a_2, a_3, ..., a_n$ and $b_1, b_2, b_3, ..., b_n$, such that $a_1 \leq a_2 \leq a_3 \leq ... \leq a_n$ and $b_1 \leq b_2 \leq b_3 \leq ... \leq b_n$, then
  $\dfrac{a_1 b_1 + a_2 b_2 + a_3 b_3 ... + a_n b_n}{n} \geq \dfrac{a_1 + a_2 + a_3 ... + a_n}{n} \times \dfrac{b_1 + b_2 + b_3 ... + b_n}{n}$.
  The inequality holds iff $a_1 = a_2 = a_3 = ... = a_n$ or $b_1 = b_2 = b_3 = ... = b_n$.

- **<u>Weierstrass' Inequality</u>**:

  (1) If $a_1, a_2, a_3, ..., a_n$ are n positive real numbers, then for $n \geq 2$
  $(1 + a_1)(1 + a_2)(1 + a_3)...(1 + a_n) > 1 + a_1 + a_2 + a_3 + ... + a_n$

  (2) If $a_1, a_2, a_3, ..., a_n$ are n positive real numbers, then for $n < 1$
  $(1 - a_1)(1 - a_2)(1 - a_3)...(1 - a_n) > 1 - a_1 - a_2 - a_3 - ... - a_n$.

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- **<u>Arithmetic Mean of *m*-th order</u>**: Let $a_1, a_2, a_3, ..., a_n$ are n positive real numbers (not all equal) and let m be a real number, then

  (1) $\dfrac{a_1^m + a_2^m + a_3^m + ... + a_n^m}{n} > \left(\dfrac{a_1 + a_2 + a_3 + ... + a_n}{n}\right)^m$ if $m \in R - [0, 1]$.

  (2) $\dfrac{a_1^m + a_2^m + a_3^m + ... + a_n^m}{n} < \left(\dfrac{a_1 + a_2 + a_3 + ... + a_n}{n}\right)^m$ if $m \in (0, 1)$.

  (3) $\dfrac{a_1^m + a_2^m + a_3^m + ... + a_n^m}{n} = \left(\dfrac{a_1 + a_2 + a_3 + ... + a_n}{n}\right)^m$ if $m \in \{0, 1\}$.

- **<u>Logarithmic Inequality</u>**:

(1) $\log_{\sim} n = \frac{1}{\ } \log_{\sim} n$.

(2) $a^{\log_c b} = b^{\log_c a}$, where $a, b, , c > 0,\ c \neq 1$.

(3) Usually $\log a^n = n \log a$, but if n is even then $\log a^n = n \log|a|$.

(4) If $a > 1,\ p > 1 \Rightarrow \log_a p > 0$.

(5) If $0 < a < 1,\ p > 1 \Rightarrow \log_a p < 0$.

(6) If $a > 1,\ 0 < p < 1 \Rightarrow \log_a p < 0$.

(7) If $p > a > 1 \Rightarrow \log_a p > 1$.

(8) If $a > p > 1 \Rightarrow 0 < \log_a p < 1$.

(9) If $0 < a < p < 1 \Rightarrow 0 < \log_a p < 1$.

(10) If $0 < p < a < 1 \Rightarrow \log_a p > 1$.

(11) If $\log_m a > b \Rightarrow \begin{cases} a > m^b & m > 1 \\ a < m^b & 0 < m < 1 \end{cases}$.

(12) If $\log_m a < b \Rightarrow \begin{cases} a < m^b & m > 1 \\ a > m^b & 0 < m < 1 \end{cases}$.

(13) If $\log_m a < \log_m b \Rightarrow \begin{cases} a \geq b & p > 1 \\ a \leq b & 0 < p < 1 \end{cases}$.

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